

Unit 7

EQUILIBRIA

44 items · 129 marks

Syllabus sections in this unit

7.1 Chemical equilibria: reversible reactions, dynamic equilibrium

7.2 Brønsted–Lowry theory of acids and bases

Unit 7: Equilibria

1. M/J 26/P21/Q2(b)(i)

Context

2 The Contact process uses sulfur to produce sulfuric acid.

stage 1 Sulfur is burned in excess air.

stage 2 $2\text{SO}_2(\text{g}) + \text{O}_2(\text{g}) \rightleftharpoons 2\text{SO}_3(\text{g}) \quad \Delta H = -196 \text{ kJ mol}^{-1}$

stage 3 SO_3 reacts to produce H_2SO_4 .

Context: part (b) (introduction)

(b) The reaction conditions for stage 2 include:

- using a ratio of 1 mol SO_2 to 1 mol O_2
- heat in the presence of V_2O_5 .

(i) Stage 2 is repeated using a ratio of more than 1 mol SO_2 to 1 mol O_2 .

Predict the effect on the percentage yield of SO_3 at equilibrium. Use Le Chatelier's Principle to explain your answer.

[1]

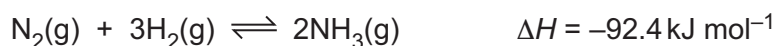
2. M/J 26/P22/Q1(e)(i)

Context

1 Oxygen reacts with many elements.

Context: part (e) (introduction)

(e) Ammonia is made in the Haber process.



A temperature of 400–450 °C is used in the Haber process.

(i) State **two** other conditions used in the Haber process.

1

2

[2]

3. M/J 26/P22/Q3(a)

also Unit 11 (1 mark); this unit 2 marks

Context

- 3** Separate samples of NaCl and NaBr are reacted with concentrated H₂SO₄. Table 3.1 shows the observations for each reaction.

Table 3.1

reaction	compound	observations
1	NaCl	- effervescence - steamy fumes that turn blue litmus paper red
2	NaBr	- effervescence - steamy fumes initially that turn blue litmus paper red - brown vapour - pungent smelling gas

- (a) (i) Write an equation for reaction 1.

[1]

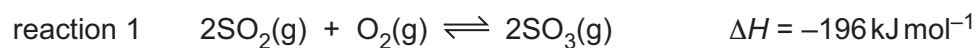
- (ii) Explain why reaction 1 is described as a Brønsted–Lowry acid–base reaction.

[2]

4. M/J 26/P23/Q3(b)

Context

- 3** The conditions used for the conversion of sulfur dioxide to sulfur trioxide in the Contact process are carefully controlled.



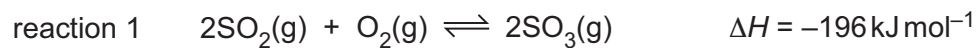
- (b) Define Le Chatelier's principle.

[2]

5. M/J 26/P23/Q3(c)

Context

- 3** The conditions used for the conversion of sulfur dioxide to sulfur trioxide in the Contact process are carefully controlled.



- (c) In industry, reaction 1 occurs at 400–450 °C and 110–200 kPa with a catalyst.

Complete Table 3.1 by using the terms 'increase', 'decrease' or 'no change'.

Table 3.1

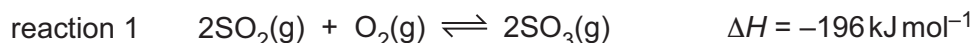
	use temperature greater than 450 °C	without catalyst
effect on value of equilibrium constant, K_p		
effect on yield of product at equilibrium		
effect on rate of reaction		

[3]

6. M/J 26/P23/Q3(d)

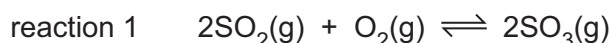
Context

- 3** The conditions used for the conversion of sulfur dioxide to sulfur trioxide in the Contact process are carefully controlled.



- (d)** A sealed tube that initially contains only 2.00 mol of $\text{SO}_2(\text{g})$ and 2.00 mol of $\text{O}_2(\text{g})$ is left to reach equilibrium.

At equilibrium, 1.96 mol of $\text{SO}_3(\text{g})$ is present and the total pressure is 150 kPa.



- (i)** Complete Table 3.2.

Table 3.2

	$\text{SO}_2(\text{g})$	$\text{O}_2(\text{g})$	$\text{SO}_3(\text{g})$
amount initially / mol	2.00	2.00	0
amount at equilibrium / mol			1.96
mole fraction at equilibrium	0.01325	0.3377	

[3]

- (ii)** Table 3.3 shows the mole fractions at equilibrium of $\text{SO}_2(\text{g})$, $\text{O}_2(\text{g})$ and $\text{SO}_3(\text{g})$ under different conditions.

Table 3.3

	$\text{SO}_2(\text{g})$	$\text{O}_2(\text{g})$	$\text{SO}_3(\text{g})$
mole fraction at equilibrium	0.360	0.019	0.621

Calculate K_p at 200 kPa using the data from Table 3.3.

Include units in your answer.

$$K_p =$$

units

[3]

7. M/J 26/P24/Q2(b)(ii)

Context: part (b) (introduction)

- (b)** Nitrogen is used to manufacture ammonia, NH_3 , in the Haber process. An iron catalyst is used.

- (ii)** State the temperature and pressure used in the Haber process.

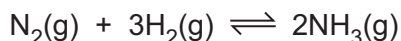
temperature =

pressure =

[2]

8. M/J 26/P24/Q2(c)

(c) The equation for the Haber process is shown.



Three separate experiments are carried out using mixtures of $\text{N}_2(\text{g})$ and $\text{H}_2(\text{g})$ to find the amount of $\text{NH}_3(\text{g})$ produced at equilibria under different conditions.

- (i) In experiment 1, 3.55 mol of $\text{N}_2(\text{g})$ and 8.60 mol of $\text{H}_2(\text{g})$ are mixed and left to reach equilibrium.

At equilibrium, 3.24 mol of NH_3 is present.

Complete Table 2.1.

Table 2.1

	$\text{N}_2(\text{g})$	$\text{H}_2(\text{g})$	$\text{NH}_3(\text{g})$
initial amount/mol	3.55	8.60	0
change in amount/mol			3.24
amount at equilibrium/mol			3.24

[3]

- (ii) In experiment 2, under different conditions $\text{N}_2(\text{g})$ and $\text{H}_2(\text{g})$ are mixed and left to reach equilibrium.

Table 2.2 shows the amount, in mol, at equilibrium of $\text{N}_2(\text{g})$, $\text{H}_2(\text{g})$ and $\text{NH}_3(\text{g})$.

Table 2.2

	$\text{N}_2(\text{g})$	$\text{H}_2(\text{g})$	$\text{NH}_3(\text{g})$
amount at equilibrium/mol	0.874	1.340	0.347

Calculate the mole fraction of $\text{NH}_3(\text{g})$ in the equilibrium mixture in experiment 2.

$$\text{mole fraction of } \text{NH}_3(\text{g}) =$$

[2]

- (iii) The expression for the equilibrium constant, K_p , for the Haber process is shown.

$$K_p = \frac{p(\text{NH}_3)^2}{p(\text{N}_2) \times p(\text{H}_2)^3}$$

Describe what is meant by partial pressure.

[1]

- (iv) In experiment 3, the mixture at equilibrium contains the following partial pressures of $\text{N}_2(\text{g})$ and $\text{H}_2(\text{g})$.

$$p(\text{N}_2) = 1590 \text{ kPa}$$

$$p(\text{H}_2) = 1750 \text{ kPa}$$

For experiment 3, the equilibrium constant, K_p , is $8.49 \times 10^{-7} \text{ kPa}^{-2}$.

Calculate the partial pressure, in kPa, of $\text{NH}_3(\text{g})$ at equilibrium.

$$p(\text{NH}_3) = \text{ kPa [3]}$$

9. MAR 26/P22/Q4(a)(i)-(iii)

Context

4 Cyanoethanoic acid, $\text{CH}_2(\text{CN})\text{COOH}$, is a weak organic acid.

Context: part (a) (introduction)

(a) At room temperature, $\text{CH}_2(\text{CN})\text{COOH}$ undergoes an acid–base reaction with $\text{NaOH}(\text{aq})$.

(i) Define weak acid.

[1]

(ii) Sketch the pH titration curve on Figure 4.1 for the reaction of:

- 25.00 cm^3 of 0.10 mol dm^{-3} $\text{CH}_2(\text{CN})\text{COOH}(\text{aq})$ and
- 0.10 mol dm^{-3} $\text{NaOH}(\text{aq})$, added to excess.

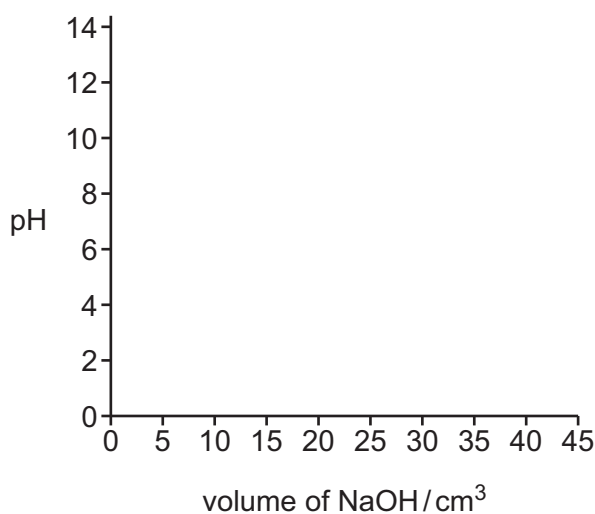


Figure 4.1

[3]

(iii) Write the overall ionic equation for the neutralisation of $\text{CH}_2(\text{CN})\text{COOH}(\text{aq})$ by $\text{NaOH}(\text{aq})$ in **4(a)(ii)**.

Include state symbols.

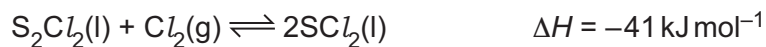
[1]

10. O/N 25/P22/Q2(e)

Context

2 The Period 3 elements show trends in physical and chemical properties across the period.

- (e) $\text{S}_2\text{Cl}_2(\text{l})$ reacts with $\text{Cl}_2(\text{g})$ in a reversible reaction to form $\text{SCl}_2(\text{l})$. Under certain conditions, a dynamic equilibrium is established.



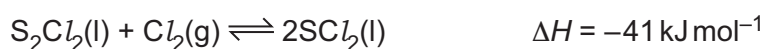
- (i) State what is meant by dynamic equilibrium.

[1]

- (ii) Identify the condition necessary to establish dynamic equilibrium.

[1]

- (iii) $\text{S}_2\text{Cl}_2(\text{l})$ is yellow and $\text{SCl}_2(\text{l})$ is red.



State what is observed when the following changes are made to an equilibrium mixture of $\text{S}_2\text{Cl}_2(\text{l})$ and $\text{SCl}_2(\text{l})$.

Explain your answers.

- The equilibrium mixture is warmed gently.
observation
explanation
- The overall pressure of the equilibrium mixture is increased.
observation
explanation

[4]

11. O/N 25/P24/Q2(b)

Context

- 2** Aluminium oxide, Al_2O_3 , and phosphorus(V) oxide, P_4O_{10} , are both used as reagents and catalysts.

(b) A 5.00 dm^3 sealed flask contains 0.400 mol of $CO(g)$ and 0.800 mol of $H_2(g)$ and an Al_2O_3 catalyst. The flask is heated to a temperature of 290°C and allowed to reach equilibrium.

Equation 1 shows the reaction.



The equilibrium constant, K_c , of equation 1 is given.

$$K_c = \frac{[CH_3OH]}{[CO][H_2]^2}$$

- (i) State the units of K_c .

[1]

- (ii) The equilibrium mixture contains 0.280 mol of $CH_3OH(g)$.

Calculate the value of K_c .

Give your answer to **three** significant figures.

value of $K_c =$ [3]

- (iii) State and explain the effect, if any, on the value of K_c when the overall pressure in the sealed flask is increased.

[1]

12. M/J 25/P21/Q3(b)

- (b) Define Le Chatelier's principle.

[2]

13. M/J 25/P21/Q3(c)

- (c) Reaction 2 shows the equilibrium reaction between X(g) and Y(g) to produce Z(g) in a sealed container.

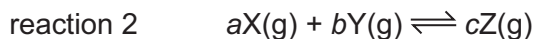


Fig. 3.3 shows the effect of changing pressure on the percentage yield of Z(g) at two different temperatures, 300 K and 350 K.

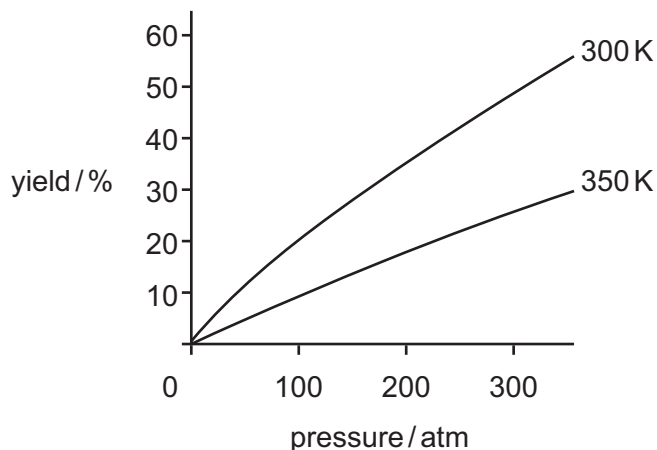


Fig. 3.3

Deduce **two** conclusions about reaction 2 using Fig 3.3.

[2]

14. M/J 25/P23/Q2(d)

Context

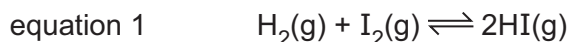
- 2** The chemical properties of oxides are related to the chemical bonding present in these compounds.
- (d) State why ZnO is described as a Brønsted–Lowry base when it is added to H₂SO₄(aq).

[1]

15. M/J 25/P24/Q3(b)

Context

- 3** H₂(g) and I₂(g) react to form HI(g) in a reversible reaction, as shown in equation 1.



In three separate experiments, a student combines different amounts of two or more of the gases from equation 1. In each experiment, the gases are left to reach equilibrium at a given temperature.

- (b) In experiment 2, 0.100 mol of H₂(g), 0.200 mol of I₂(g) and 0.300 mol of HI(g) are combined.

At equilibrium 0.154 mol of I₂ is present.

Calculate the amounts, in mol, of H₂ and HI in the equilibrium mixture produced in experiment 2.

amount of H₂ in equilibrium mixture = _____ mol

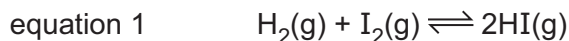
amount of HI in equilibrium mixture = _____ mol

[3]

16. M/J 25/P24/Q3(c)

Context

- 3** $\text{H}_2(\text{g})$ and $\text{I}_2(\text{g})$ react to form $\text{HI}(\text{g})$ in a reversible reaction, as shown in equation 1.



In three separate experiments, a student combines different amounts of two or more of the gases from equation 1. In each experiment, the gases are left to reach equilibrium at a given temperature.

- (c)** In experiment 3, 0.0772 mol of H_2 and 0.0986 mol of I_2 are present in an equilibrium mixture at 298 K.



Use the equilibrium constant, K_{c} , to calculate the amount, in mol, of HI present in the equilibrium mixture in experiment 3.

amount of HI in equilibrium mixture = _____ mol [1]

17. O/N 24/P21/Q3(b)(iii)

Context

- 3** The Group 14 elements show a change from non-metallic to metallic character down the group.

Context: part (b) (introduction)

- (b)** Carbon is found in inorganic compounds such as carbonates.

- (iii)** Ammonium carbonate undergoes an acid–base reaction with $\text{NaOH}(\text{aq})$.

Explain this statement.

[2]

18. O/N 24/P23/Q3(b)(iii)

Context

- 3** The Group 14 elements show a change from non-metallic to metallic character down the group.

Context: part (b) (introduction)

- (b)** Carbon is found in inorganic compounds such as carbonates.

- (iii)** Ammonium carbonate undergoes an acid–base reaction with $\text{NaOH}(\text{aq})$.

Explain this statement.

[2]

19. M/J 24/P21/Q2(c)

also Unit 6 (1 mark), Unit 10 (1 mark); this unit 2 marks

(c) Solid magnesium nitride, Mg_3N_2 , is a crystalline solid.

- (i) Deduce the oxidation numbers of magnesium and nitrogen in magnesium nitride to complete Table 2.1.

Table 2.1

	oxidation number in Mg_3N_2
magnesium	
nitrogen	

[1]

- (ii) Magnesium nitride reacts with an excess of water to produce ammonia and magnesium hydroxide only. Construct an equation to describe this reaction.

[1]

- (iii) Explain why the solution produced in the reaction in (c)(ii) has a pH greater than 7. Refer to the products of the reaction in your answer.

[2]

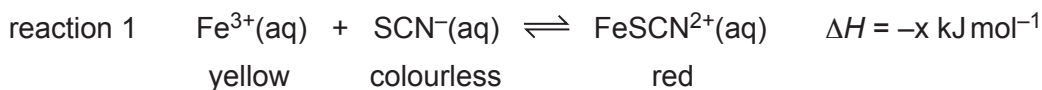
20. M/J 24/P21/Q3(a)

- 3 (a) Define Le Chatelier's principle.

[2]

21. M/J 24/P21/Q3(b)-(c)

- (b) Reaction 1 describes the reversible reaction between yellow $\text{Fe}^{3+}(\text{aq})$ and colourless $\text{SCN}^{-}(\text{aq})$ to produce red $\text{FeSCN}^{2+}(\text{aq})$.



A mixture of $\text{Fe}^{3+}(\text{aq})$, $\text{SCN}^{-}(\text{aq})$ and $\text{FeSCN}^{2+}(\text{aq})$ is at equilibrium at 20°C .

The temperature of this mixture is then increased to 50°C and allowed to reach equilibrium.

Deduce the changes that occur, if any, in the equilibrium mixture at 50°C compared to the equilibrium mixture at 20°C .

- change in appearance
- change in relative concentration of $\text{FeSCN}^{2+}(\text{aq})$
- change in value of the equilibrium constant, K_c

[3]

- (c) In another experiment, equimolar amounts of $\text{Fe}^{3+}(\text{aq})$ and $\text{SCN}^{-}(\text{aq})$ are mixed together and allowed to reach equilibrium. The total volume of the mixture is 25.0 cm^3 .



At equilibrium the mixture contains:

- $[\text{SCN}^{-}] = 1.30 \times 10^{-3} \text{ mol dm}^{-3}$
- $[\text{FeSCN}^{2+}] = 0.300 \times 10^{-3} \text{ mol dm}^{-3}$.

- (i) Calculate the initial amount, in mol, of $\text{Fe}^{3+}(\text{aq})$ added to $\text{SCN}^{-}(\text{aq})$ to produce this mixture.

initial amount of $\text{Fe}^{3+}(\text{aq}) =$ mol [2]

- (ii) Calculate K_c for reaction 1 and state its units.

Show your working.

 $K_c =$

units

[2]

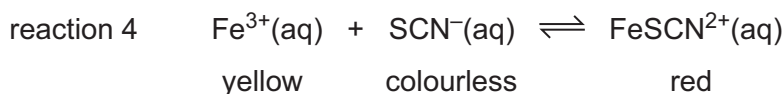
22. M/J 24/P22/Q3(a)

- 3 (a) Describe what is meant by dynamic equilibrium.

[2]

23. M/J 24/P22/Q3(b)

- (b) Reaction 4 describes the reversible reaction between yellow $\text{Fe}^{3+}(\text{aq})$ and colourless $\text{SCN}^{-}(\text{aq})$ to produce red $\text{FeSCN}^{2+}(\text{aq})$.



An equilibrium mixture contains $\text{Fe}^{3+}(\text{aq})$, $\text{SCN}^{-}(\text{aq})$ and $\text{FeSCN}^{2+}(\text{aq})$. A few colourless crystals of soluble $\text{KSCN}(\text{s})$ are added. The mixture is then left until it reaches equilibrium again. The temperature of both equilibrium mixtures is the same.

- (i) Deduce the changes that occur, if any, in the equilibrium mixture after $\text{KSCN}(\text{s})$ is added compared to the original equilibrium mixture.
- change in appearance
 - change in relative concentration of $\text{Fe}^{3+}(\text{aq})$
 - change in value of the equilibrium constant, K_c

[3]



7

- (ii) The expression for the equilibrium constant, K_c , for reaction 4 is shown.

$$K_c = \frac{[\text{FeSCN}^{2+}(\text{aq})]}{[\text{Fe}^{3+}(\text{aq})] \times [\text{SCN}^{-}(\text{aq})]}$$

$5.00 \times 10^{-5} \text{ mol}$ of $\text{Fe}^{3+}(\text{aq})$ and $5.00 \times 10^{-5} \text{ mol}$ of $\text{SCN}^{-}(\text{aq})$ are added together and allowed to reach equilibrium. The total volume of the mixture is 25.0 cm^3 .

At equilibrium the concentration of $\text{FeSCN}^{2+}(\text{aq})$ is $4.23 \times 10^{-4} \text{ mol dm}^{-3}$.

Calculate the equilibrium constant, K_c , for reaction 4.

Include the units in your answer.

 $K_c =$

units

[4]



8

24. M/J 24/P23/Q2(a)

Context

- 2** $\text{N}_2(\text{g})$ reacts with $\text{H}_2(\text{g})$ in the Haber process, as shown in reaction 1.

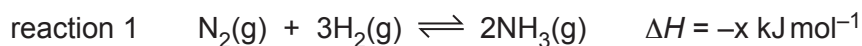


Table 2.1 shows the different conditions used to produce three equilibrium mixtures, **A**, **B** and **C**.

Table 2.1

	A	B	C
initial molar ratio of $\text{N}_2 : \text{H}_2$ added	1 : 3	1 : 3	1 : 3
temperature / °C	500	500	1000
pressure / atm	1000	1000	1000
iron present in mixture	no	yes	no
percentage yield of $\text{NH}_3(\text{g})$ at equilibrium	58	x	y

- (a) Describe and explain the change, if any, to the percentage yield of $\text{NH}_3(\text{g})$ produced in **B** compared to **A**.

[1]

25. M/J 24/P23/Q2(b)(i)

Context

- 2** $\text{N}_2(\text{g})$ reacts with $\text{H}_2(\text{g})$ in the Haber process, as shown in reaction 1.

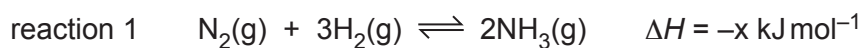


Table 2.1 shows the different conditions used to produce three equilibrium mixtures, **A**, **B** and **C**.

Table 2.1

	A	B	C
initial molar ratio of $\text{N}_2 : \text{H}_2$ added	1 : 3	1 : 3	1 : 3
temperature / °C	500	500	1000
pressure / atm	1000	1000	1000
iron present in mixture	no	yes	no
percentage yield of $\text{NH}_3(\text{g})$ at equilibrium	58	x	y

- (b) (i) Describe and explain the change, if any, to the percentage yield of $\text{NH}_3(\text{g})$ produced in **C** compared to **A**.

[1]

26. M/J 24/P23/Q2(c)

Context

- 2** $\text{N}_2(\text{g})$ reacts with $\text{H}_2(\text{g})$ in the Haber process, as shown in reaction 1.

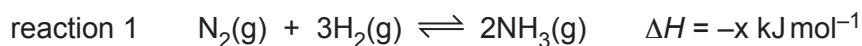


Table 2.1 shows the different conditions used to produce three equilibrium mixtures, **A**, **B** and **C**.

Table 2.1

	A	B	C
initial molar ratio of $\text{N}_2 : \text{H}_2$ added	1 : 3	1 : 3	1 : 3
temperature / °C	500	500	1000
pressure / atm	1000	1000	1000
iron present in mixture	no	yes	no
percentage yield of $\text{NH}_3(\text{g})$ at equilibrium	58	x	y

- (c) (i) Write an expression for the equilibrium constant, K_p , for reaction 1. State the units.

$$K_p =$$

units

[2]

- (ii) Equilibrium mixture **D** is made when 1.0 mol of $\text{N}_2(\text{g})$ and 3.0 mol of $\text{H}_2(\text{g})$ are added to a sealed container at 750 °C and 1000 atm and left to reach equilibrium. This mixture contains 1.16 mol of $\text{NH}_3(\text{g})$.

Calculate the mole fraction of $\text{NH}_3(\text{g})$ in **D**.

$$\text{mole fraction of } \text{NH}_3(\text{g}) =$$

[2]

- (iii) The mole fraction of $\text{N}_2(\text{g})$ is 0.625 in a new equilibrium mixture, **E**.

Calculate the partial pressure of $\text{N}_2(\text{g})$ in **E** when the total pressure is 1000 atm.

$$\text{partial pressure of } \text{N}_2(\text{g}) =$$

atm [1]

27. M/J 24/P23/Q4(b)(i)

Context: part (b) (introduction)

- (b) An exothermic reaction occurs when $\text{NH}_4^+(\text{aq})$ is added to $\text{OH}^-(\text{aq})$.

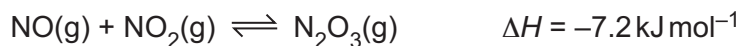
- (i) Identify the type of reaction.

[1]

28. O/N 23/P22/Q2(a)

Context

- 2** NO and NO₂ react at 25 °C to give N₂O₃ as shown in the equation.



The reaction is reversible and reaches equilibrium in a closed system.

- (a)** Fig. 2.1 shows how the rate of the forward reaction changes with time.

Initially, the rate of the reverse reaction is zero.

Complete Fig. 2.1 to sketch how the rate of the **reverse** reaction changes with time.

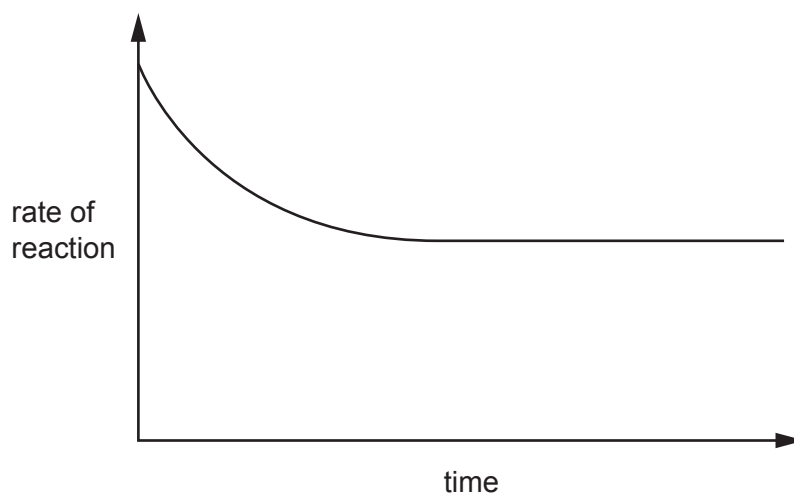


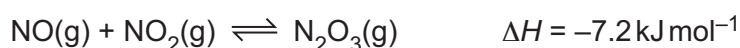
Fig. 2.1

[1]

29. O/N 23/P22/Q2(b)

Context

- 2** NO and NO₂ react at 25 °C to give N₂O₃ as shown in the equation.



The reaction is reversible and reaches equilibrium in a closed system.

- (b)** State how the position of equilibrium changes, if at all, when the reaction takes place at 100 °C.

Explain your answer.

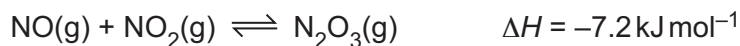
Assume the pressure remains constant.

[2]

30. O/N 23/P22/Q2(c)

Context

- 2** NO and NO₂ react at 25 °C to give N₂O₃ as shown in the equation.



The reaction is reversible and reaches equilibrium in a closed system.

- (c)** Table 2.1 shows the composition of an equilibrium mixture of NO(g), NO₂(g) and N₂O₃(g) at 101 kPa.

Table 2.1

gas	number of moles at equilibrium / mol
NO	0.605
NO ₂	0.605
N ₂ O ₃	0.390

Calculate K_p , the equilibrium constant with respect to partial pressures.

Deduce the units of K_p .

$K_p =$ units

[3]

31. O/N 23/P22/Q3(a)(ii)

Context

- 3** Phosphoric(V) acid, H₃PO₄, is used in both inorganic and organic reactions.

Context: part (a) (introduction)

- (a)** H₃PO₄ is made in a two-step process from phosphorus.

step 1 Phosphorus reacts with an excess of oxygen to form a white solid.

step 2 The white solid then reacts with water to form H₃PO₄.

- (ii)** H₃PO₄ is a weak Brønsted–Lowry acid.

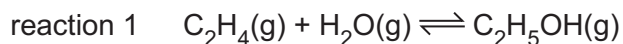
Define weak Brønsted–Lowry acid.

[2]

32. M/J 23/P22/Q4(b)

Context

- 4** In industry, ethanol is made by reacting ethene with steam in the presence of H_3PO_4 .



- (b)** Reaction 1 reaches equilibrium at constant temperature and pressure.

Deduce what effect increasing the pressure will have on the amount of ethanol in the new equilibrium mixture. Use Le Chatelier's principle to explain your answer.

effect of increasing pressure

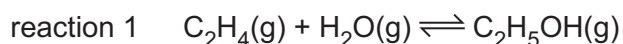
explanation

[2]

33. M/J 23/P22/Q4(c)(i)

Context

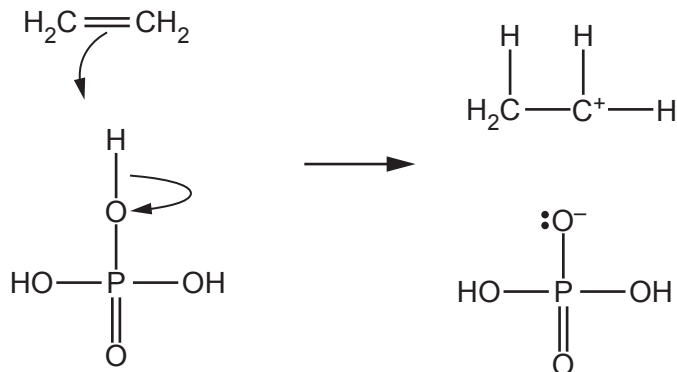
- 4** In industry, ethanol is made by reacting ethene with steam in the presence of H_3PO_4 .



Context: part (c) (introduction)

- (c)** The mechanism for reaction 1 can be described in three steps. Steps 1 and 2 for reaction 1 are shown in Fig. 4.1.

step 1



step 2

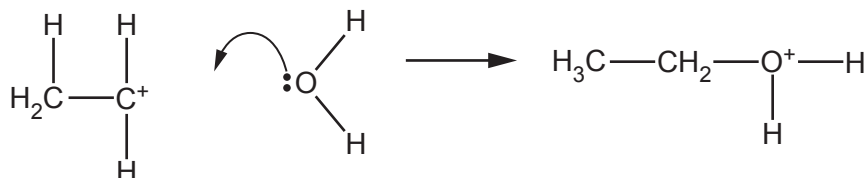


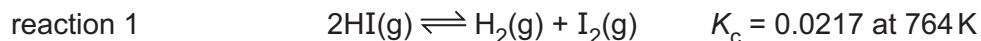
Fig. 4.1

- (i)** Describe the behaviour of H_3PO_4 in step 1 in Fig. 4.1. Explain your answer.

[1]

34. M/J 23/P23/Q4(b)

- (b) A sample of HI(g) is added to a 2.00 dm³ sealed vessel at 764 K and allowed to reach equilibrium.



At equilibrium the mixture contains 1.70 mol of HI(g).

- (i) State **one** difference in the appearance of the initial reaction mixture compared to the mixture at equilibrium.

[1]

- (ii) Deduce the expression for equilibrium constant K_c for reaction 1.

$$K_c =$$

[1]

- (iii) Calculate the concentration of I₂ present in the reaction mixture at equilibrium.
Show your working.

concentration of I₂ = mol dm⁻³ [3]

35. M/J 23/P23/Q4(c)

- (c) The experiment is repeated at 500 K. The value of K_c under these conditions is 0.00625.

- (i) Describe the difference in the composition of the equilibrium mixture at 500 K compared to 764 K.

[1]

- (ii) Use Le Chatelier's principle to deduce whether the decomposition of HI(g) is endothermic or exothermic. Explain your answer.

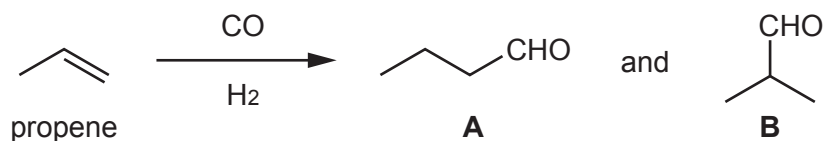
[1]

36. MAR 23/P22/Q3(d)(ii)

Context

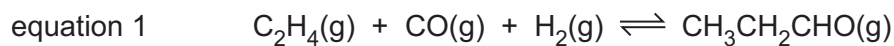
- 3** Alkenes undergo an addition reaction with a 1:1 mixture of CO and H₂ to form aldehydes.

Fig. 3.1 shows the reaction of propene with a 1:1 mixture of CO and H₂.

**Fig. 3.1**

Context: part (d) (introduction)

- (d)** The reaction of ethene, C₂H₄, with a 1:1 mixture of CO and H₂ is shown in equation 1.



At atmospheric pressure a cobalt-based catalyst is used in this reaction.

- (ii)** Explain why the yield of CH₃CH₂CHO(g) increases when the overall pressure of the reaction mixture is increased.

[1]

37. O/N 22/P22/Q1(c)(i)

Context

- 1** Species such as NH₄⁺, CO₃²⁻ and PO₄³⁻ are examples of molecular ions.

Context: part (c) (introduction)

- (c)** NH₄⁺ is a Brønsted–Lowry acid.

- (i)** Define Brønsted–Lowry acid.

[1]

38. O/N 22/P22/Q2(c)(ii)

Context

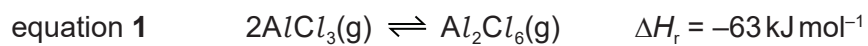
2 The chlorides of some of the Period 3 elements are shown in Table 2.1.

Table 2.1

Period 3 chloride	NaCl	AlCl ₃	SiCl ₄	PCl ₅	PCl ₃	SCl ₂
bonding					C	C
structure					S	S
oxidation state of Period 3 element						

Context: part (c) (introduction)

(c) In the gas phase, AlCl₃(g) exists at equilibrium with Al₂Cl₆(g) as shown.



(ii) State the effect of an increase in temperature on the equilibrium mixture in equation 1. Explain your answer.

[1]

39. M/J 22/P21/Q2(d)

Context

2 Nitrogen molecules, $\text{N}_2(\text{g})$, contain two atoms attracted to each other by a triple covalent bond.

(d) 25 cm^3 of $0.10\text{ mol dm}^{-3}\text{ HCl}(\text{aq})$ is added to a beaker and its pH is recorded.

50 cm^3 of $0.10\text{ mol dm}^{-3}\text{ NH}_3(\text{aq})$ is added to the $\text{HCl}(\text{aq})$ in 5 cm^3 portions.

The pH of the mixture is monitored until all the $\text{NH}_3(\text{aq})$ is added.

HCl is a strong Brønsted-Lowry acid.

(i) Describe what is meant by a strong Brønsted-Lowry acid.

[2]

(ii) NH_3 is a weak base.

Construct an equation that shows the behaviour of NH_3 as a weak Brønsted-Lowry base when dissolved in water.

[1]

(iii) On Fig. 2.1 sketch a graph to show the change in pH which occurs when $\text{HCl}(\text{aq})$ is titrated with $\text{NH}_3(\text{aq})$ as described in **(d)**.

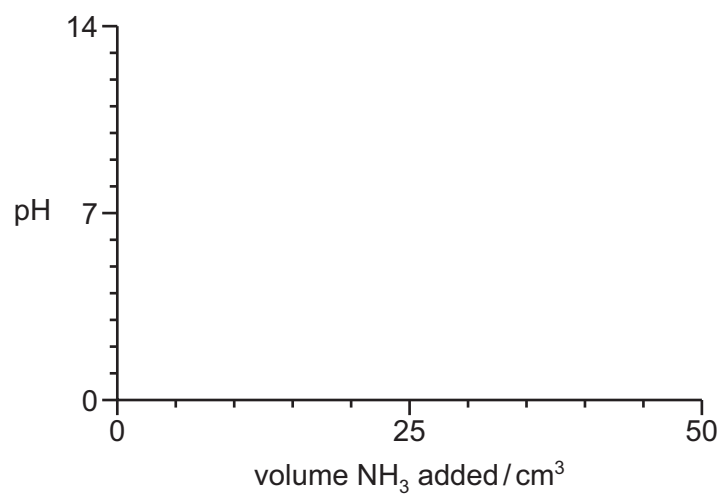
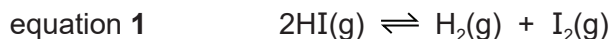


Fig. 2.1

[2]

40. M/J 22/P22/Q3(a)

- 3 (a)** 0.025 mol of HI(g) is added to a closed vessel and left to reach dynamic equilibrium. The total pressure of the vessel is 100 kPa.



- (i)** Explain what is meant by dynamic equilibrium.

[2]

- (ii)** Describe **one** difference in the initial appearance of the reaction mixture compared to the mixture at equilibrium.

[1]

- (iii)** Write an expression for K_p for the reaction described in equation 1.

$$K_p =$$

[1]

- (iv)** At equilibrium the partial pressure of HI(g) is 86.4 kPa.

Calculate the amount of HI(g) present in the mixture at equilibrium. Show your working.

amount of HI(g) = mol [2]

41. M/J 22/P22/Q3(c)

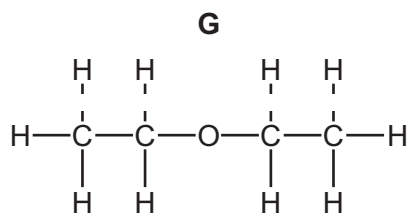
- (c)** Describe the effect of increasing pressure on the value of K_p for the decomposition of HI(g).

[1]

42. M/J 22/P23/Q3(a)(iii)

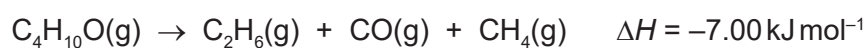
Context

3 **G** belongs to a group of compounds called ethers.

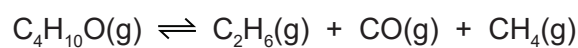
**Fig. 3.1**

Context: part (a) (introduction)

(a) When **G**, $\text{C}_4\text{H}_{10}\text{O}$, is heated, thermal decomposition occurs.



(iii) When **G**, $\text{C}_4\text{H}_{10}\text{O}$, is heated in a sealed container, an equilibrium mixture is produced.



Complete the expression for the equilibrium constant, K_c , for this reaction.
State the units of K_c .

 $K_c =$

units =

[2]

43. MAR 22/P22/Q2(d)(iii)

Context

2 Some oxides of elements in Period 3 are shown.



Context: part (d) (introduction)

(d) SO_2 and SO_3 are found in the atmosphere.

The oxidation of SO_2 to SO_3 in the atmosphere is catalysed by NO_2 .

The first step of the catalytic oxidation is shown in equation 1.



(iii) Fig. 2.2 shows how the temperature of the atmosphere varies with height from the ground.

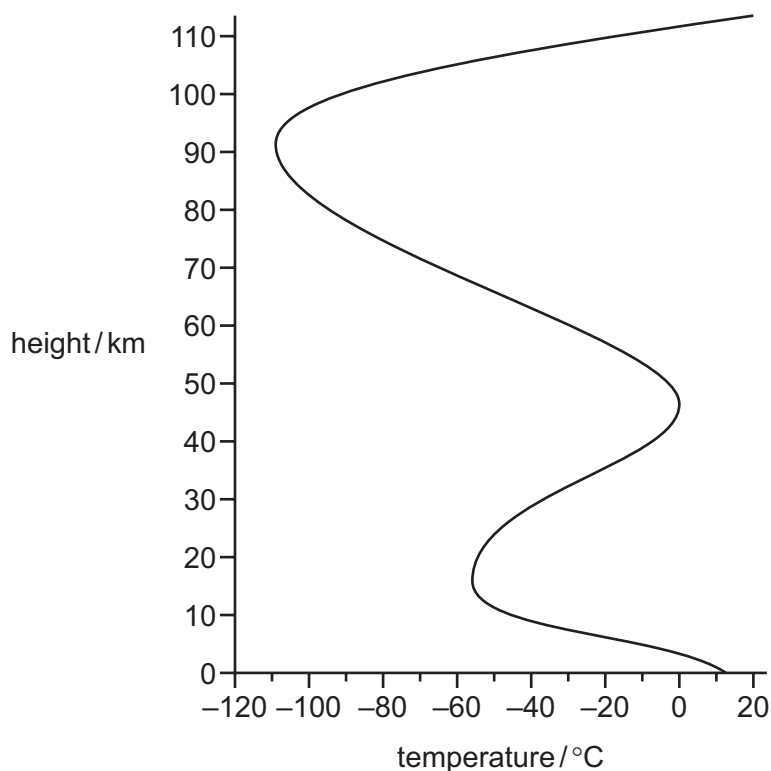


Fig. 2.2

The equilibrium reaction in equation 1 has $\Delta H_r = -168 \text{ kJ mol}^{-1}$.

Suggest how the position of this equilibrium differs at a height of 20 km compared with a height of 50 km from the ground.

Explain your answer.

[2]

44. MAR 22/P22/Q3(d)

Context

3 The hydrogen halides HCl , HBr and HI are all colourless gases at room temperature.

(d) The hydrogen halides dissolve in water to form strong Brønsted–Lowry acids.

The concentration of a strong acid can be determined by titration.

(i) State what is meant by strong Brønsted–Lowry acid.

[2]

(ii) On Fig. 3.2, sketch the pH titration curves produced when:

- $0.1 \text{ mol dm}^{-3} \text{ NaOH(aq)}$ is added to 25 cm^3 of $0.1 \text{ mol dm}^{-3} \text{ HBr(aq)}$, to excess
- $0.1 \text{ mol dm}^{-3} \text{ NH}_3(\text{aq})$ is added to 25 cm^3 of $0.1 \text{ mol dm}^{-3} \text{ HBr(aq)}$, to excess.

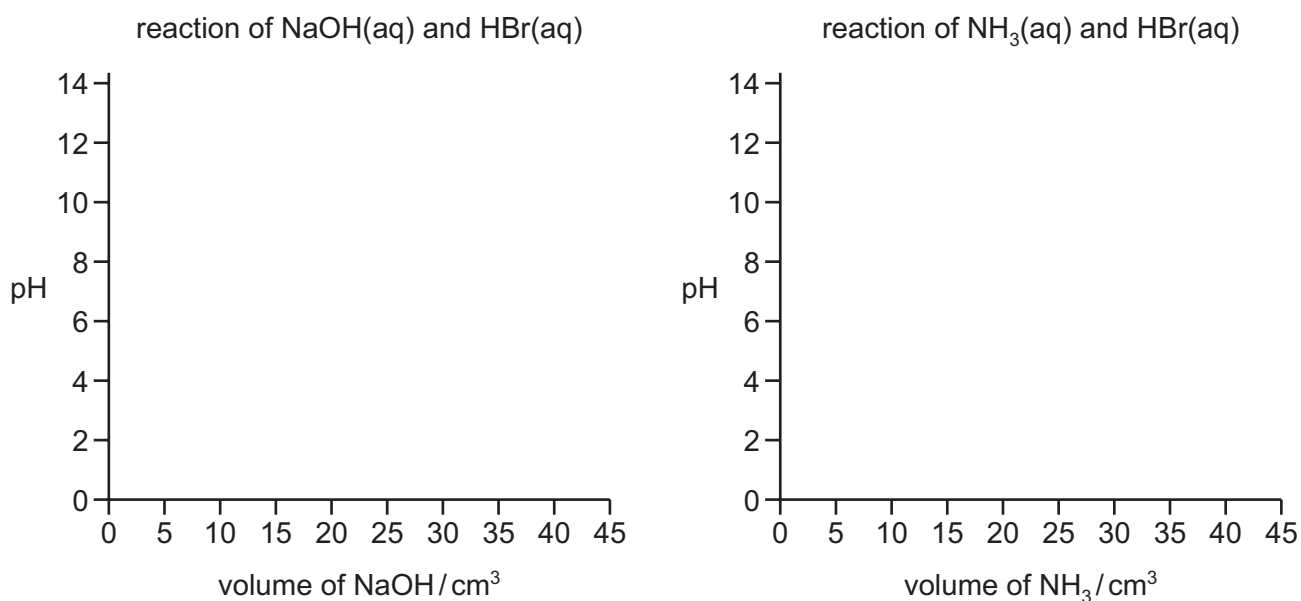


Fig. 3.2

[3]

Answers Section

1. M/J 26/P21/Q2(b)(i)

2(b)(i)	increases yield AND <u>equilibrium</u> shifts to right (hand side)	1 A alternatives to increased yield = increased amount / more product / SO_3 A <u>equilibrium</u> shifts to the side with fewer gaseous molecules / to side with fewer gaseous moles A <u>equilibrium</u> shifts to the right to reduce the concentration of SO_2 BOD <u>equilibrium</u> shifts to (favour) forwards reaction / direction I equilibrium shifts to side with lower volume I limiting reagent I any comments on rate R equilibrium shifts to side with more gaseous moles
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2. M/J 26/P22/Q1(e)(i)

1(e)(i)	M1 high pressure OR ≥ 200 atm M2 iron catalyst / Fe catalyst	2 A answers on either line M1 BOD equivalent pressure of ≥ 150 atm or equivalent in alternative units with appropriate conversion M1 A $\geq 15\,000$ kPa / ≥ 150 bar M1 I ATP M1 I numbers with no units M2 A alternative correct named catalysts e.g. Co (consult TL if necessary) M2 BOD Fe(II) catalyst / Fe^{2+} catalyst / Fe^{3+} catalyst / iron oxide catalyst M2 BOD magnetite catalyst M2 I Ni M2 I promoters M2 R Pt / platinum / vanadium oxide
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3. M/J 26/P22/Q3(a)

3(a)(i)	$\text{H}_2\text{SO}_4 + \text{NaCl} \rightarrow \text{NaHSO}_4 + \text{HCl}$ OR $\text{H}_2\text{SO}_4 + 2\text{NaCl} \rightarrow \text{Na}_2\text{SO}_4 + 2\text{HCl}$	1 A $\text{Cl}^- + \text{H}_2\text{SO}_4 \rightarrow \text{HCl} + \text{HSO}_4^-$ A $2\text{Cl}^- + \text{H}_2\text{SO}_4 \rightarrow 2\text{HCl} + \text{SO}_4^{2-}$
3(a)(ii)	M1 H_2SO_4 acid AND Cl^- base M2 H_2SO_4 donates proton(s) AND Cl^- accepts protons OR H_2SO_4 gives / transfers protons <u>to Cl^-</u>	2 proton = hydrogen ion I general definition of BL acids and bases. $\text{Cl}^- = \text{Cl}$ in NaCl I HCl is base/protons given to HCl Alternative if neither M1 nor M2 can be awarded: • H_2SO_4 acid • Cl^- base • H_2SO_4 donates proton(s) • Cl^- accepts proton(s) •✓•✓

4. M/J 26/P23/Q3(b)

3(b)	M1 (if a change in conditions occurs) the equilibrium shifts M2 to minimise the change (in conditions)	2 M1 A equilibrium moves / equilibrium changes M1 I there will be a 'change in the system' M2 minimise = oppose = counteract = resists = reverse
------	--	--

5. M/J 26/P23/Q3(c)

3(c)		use temp. greater than 450°C	without a catalyst	3	All independent including K_p and yield marks
	effect on value of equilibrium constant, K_p	decreases	no change		
	effect on yield of product at equilibrium	decreases	no change		
	effect on rate of reaction	increases	decreases		

•✓•✓•✓

6. M/J 26/P23/Q3(d)

3(d)(i)		SO ₂ (g)	O ₂ (g)	SO ₃ (g)	3	Independent marks If M3 not 0.649 look to see if ecf from M1 and / or M2 M3 A 0.649 / 0.65 M3 A ecf from M1 and / or M2 M3 A 1 – (0.01325 + 0.3377) = 0.649(05)
	amount initially / mol	2	2	0		
	amount at equilibrium / mol	M1 0.04(0)	M2 (2–0.98) = 1.02	1.96		
	mol fraction at equilibrium	0.01325	0.3377	M3 (1.96 / 3.02) = 0.649(007)		

3(d)(ii)	$\frac{(0.621 \times 200)(0.621 \times 200)}{(0.360 \times 200)(0.360 \times 200)(0.019 \times 200)} = 0.783(1) \text{ kPa}^{-1}$ OR $7.83(1) \times 10^{-4} \text{ Pa}^{-1}$ ✓✓✓ M1 correct expression for K_p using numerical values M2 correct value for K_p. M3 kPa^{-1}	3	0.783(1) kPa^{-1} ✓✓✓ $7.83(1) \times 10^{-4} \text{ Pa}^{-1}$ ✓✓✓ M1 I use of [] for () M2 A ecf from M1 for transcription errors of mol fraction values M2 A ecf from M1 for inverted expression i.e. $\frac{0.360 \times 200)(0.360 \times 200)(0.019 \times 200)}{(0.621 \times 200)(0.621 \times 200)} = 1.27(7)$ AND M3 kPa M3 A kPa^{-1} if no expression / incorrect K_p expression M3 A ecf for appropriate units based on incorrect expression for K_p M3 A 'no unit' if expression used involved no reference to 200kPa and only mol fraction (= 156.6) M3 A Pa^{-1} if calculation is based on conversion of kPa to Pa i.e. $7.83(1) \times 10^{-4} \text{ Pa}^{-1}$ ✓✓✓ I 1 / kPa I $K_p = (\text{ppSO}_3)^2 \div (\text{ppSO}_2)^2 \times (\text{ppO}_2)$
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7. M/J 26/P24/Q2(b)(ii)

2(b)(ii)	M1 (temperature:) 400–450 °C	2	Units required M1 A alternative temperatures with appropriate units A 400–500 °C A 673–773 K M2 A 15 200–25 300 kPa A 15 200 000–25 300 000 Pa A 150–250 bar
	M2 (pressure:) 150–250 atm		

8. M/J 26/P24/Q2(c)

2(c)(i)	<table><tr><td></td><td>N₂(g)</td><td>H₂(g)</td><td>NH₃(g)</td></tr><tr><td>initial amount / mol</td><td>3.55</td><td>8.60</td><td>0</td></tr><tr><td>change in amount / mol</td><td>(-) 1.62</td><td>(-) 4.86</td><td>3.24</td></tr><tr><td>amount at equilibrium / mol</td><td>(+) 1.93</td><td>(+) 3.74</td><td>3.24</td></tr></table>		N ₂ (g)	H ₂ (g)	NH ₃ (g)	initial amount / mol	3.55	8.60	0	change in amount / mol	(-) 1.62	(-) 4.86	3.24	amount at equilibrium / mol	(+) 1.93	(+) 3.74	3.24	3 M1 (-) 1.62 M2 (-) 4.86 M3 N₂ and H₂ amount / moles at equilibrium (initial amount / moles minus change in amount / moles) M3 A ecf from incorrect changes in amount / moles in M1 / M2 M3 R negative amounts / moles at equilibrium Max 2: <table><tr><td></td><td>N₂(g)</td><td>H₂(g)</td><td>NH₃(g)</td></tr><tr><td>initial amount / mol</td><td>3.55</td><td>8.60</td><td>0</td></tr><tr><td>change in amount / mol</td><td>(-)1.93</td><td>(-)3.74</td><td>3.24</td></tr><tr><td>amount at equilibrium /mol</td><td>(+)1.62</td><td>(+)4.86</td><td>3.24</td></tr></table>		N ₂ (g)	H ₂ (g)	NH ₃ (g)	initial amount / mol	3.55	8.60	0	change in amount / mol	(-)1.93	(-)3.74	3.24	amount at equilibrium /mol	(+)1.62	(+)4.86	3.24
	N ₂ (g)	H ₂ (g)	NH ₃ (g)																															
initial amount / mol	3.55	8.60	0																															
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amount at equilibrium / mol	(+) 1.93	(+) 3.74	3.24																															
	N ₂ (g)	H ₂ (g)	NH ₃ (g)																															
initial amount / mol	3.55	8.60	0																															
change in amount / mol	(-)1.93	(-)3.74	3.24																															
amount at equilibrium /mol	(+)1.62	(+)4.86	3.24																															
2(c)(ii)	M1 Total amount / mol = 2.561 OR 0.874 + 1.340 + 0.347 M2 Mole fraction of NH₃ = 0.347 ÷ 2.561 = 0.135	2 R M1 and M2 if 0.874 + 1.340 + 0.347 is not used to calculate total amount / moles e.g. total moles = 0.874 + 1.340 M2 A ecf if 0.874 + 1.340 + 0.347 are used to calculate total amount / moles, but incorrectly e.g. 0.874 + 1.340 + 0.347² A correctly rounded to minimum of 2 significant figures A as integer fraction e.g. 347 / 2561 I 13.5% A calculator answer 0.135494																																
2(c)(iii)	(partial pressure) = mole fraction (of a gas) × total pressure OR the pressure that would be exerted by (only) one of the gases in a mixture (of gases) (if it occupied the same volume)	1 A product of mole fraction and total pressure A the pressure that is exerted by one of the gases to the total pressure A The pressure that a gas in a mixture of gases exerts on the surroundings A ratio / proportion / fraction of the pressure of a gas to the total pressure (in a closed system) A gas = molecules = particles R compound / element I amount of pressure that a particular gas occupies compared to total pressure																																
2(c)(iv)	M1 $p(\text{NH}_3)^2 = K_p \times p(\text{N}_2) \times p(\text{H}_2)^3$ OR $p(\text{NH}_3)^2 = 8.49 \times 10^{-7} \times 1590 \times (1750)^3$ OR $p(\text{NH}_3) = \sqrt{8.49 \times 10^{-7} \times 1590 \times (1750)^3}$ M2 $p(\text{NH}_3)^2 = 7.23(4673906) \times 10^6$ M3 $\sqrt{\text{M2}} = 2690 / 2689.73$	3 M2 ecf from.... M1 transcription error or M1 incorrect rearrangement for $p(\text{NH}_3)^2$ using all the terms K_p AND $p(\text{N}_2)$ AND $p(\text{H}_2)^3$ M2 A calculator answer 723 4673.906 M3 ecf $\sqrt{\text{M2}}$ correctly calculated to minimum 3 significant figures M3 A calculator answer 2689.73																																

9. MAR 26/P22/Q4(a)(i)-(iii)

4(a)(i)	(proton donor that) partially dissociates (in aqueous solution)	1
4(a)(ii)	M1 weak acid curve starts pH 1.7 (calculation not necessary) M2 vertical section at 25 cm ³ M3 strong alkali curve ends at pH 13	3
4(a)(iii)	$\text{H}^+(\text{aq}) + \text{OH}^-(\text{aq}) \rightarrow \text{H}_2\text{O}(\text{l})$	1

10. O/N 25/P22/Q2(e)

2(e)(i)	rate of forward and backward (reaction) are the same / equal OR (ratio of) concentrations of reactants and products are constant	1
2(e)(ii)	closed system	1
2(e)(iii)	M1 (mixture / solution becomes more) yellow M2 (forward) reaction is exothermic (equilibrium shifts left) OR equilibrium / reaction shifts (left) to absorb (additional) heat / energy M3 (mixture / solution becomes more) red M4 fewer moles of gas on right-hand side OR (gas on left and) no gas on the right	4

11. O/N 25/P24/Q2(b)

2(b)(i)	mol ⁻² dm ⁶	1
2(b)(ii)	M1 at equilibrium: moles of CO = 0.120 AND moles of H ₂ = 0.240 M2 [CH ₃ OH] = 0.056 AND [CO] = 0.024 AND [H ₂] = 0.048 M3 use of K_c with correct evaluation = 1.01×10^3	3
2(b)(iii)	no effect AND K_c is not affected by pressure	1

12. M/J 25/P21/Q3(b)

3(b)	M1 (if a change is made at a system is dynamic equilibrium) the equilibrium moves M2 to minimise the change in conditions	2
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13. M/J 25/P21/Q3(c)

3(c)	(forward) reaction is exothermic (a + b) > c	2
------	---	----------

14. M/J 25/P23/Q2(d)

2(d)	it / ZnO accepts protons	1
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15. M/J 25/P24/Q3(b)

3(b)	M1 $0.2 - 0.154 = 0.046$ (mol H ₂ and / or I ₂ reacts) M2 mol H ₂ at equilibrium $0.1 - 0.046 = 0.054$ M3 mol HI at equilibrium $0.3 + (0.046 \times 2) = 0.392$	3
------	--	----------

16. M/J 25/P24/Q3(c)

3(c)	2.46	1
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17. O/N 24/P21/Q3(b)(iii)

3(b)(iii)	M1 NH_4^+ / ammonium ion / ammonium cation is a proton / H^+ donor M2 OH^- is a proton / H^+ acceptor OR the OH^- / NaOH is a stronger base than NH_3	2
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18. O/N 24/P23/Q3(b)(iii)

3(b)(iii)	M1 NH_4^+ / ammonium ion / ammonium cation is a proton / H^+ donor M2 OH^- is a proton / H^+ acceptor OR the OH^- / NaOH is a stronger base than NH_3	2
-----------	---	----------

19. M/J 24/P21/Q2(c)

2(c)(i)	<table><tr><td></td><td>oxidation number in Mg₃N₂</td></tr><tr><td>magnesium</td><td>(+)II / 2(+)</td></tr><tr><td>nitrogen</td><td>–III/ 3 –</td></tr></table> Both correct for one mark		oxidation number in Mg ₃ N ₂	magnesium	(+)II / 2(+)	nitrogen	–III/ 3 –	1
	oxidation number in Mg ₃ N ₂							
magnesium	(+)II / 2(+)							
nitrogen	–III/ 3 –							
2(c)(ii)	Mg ₃ N ₂ + 6H ₂ O → 3Mg(OH) ₂ + 2NH ₃	1						
2(c)(iii)	M1 due to OH [–] (aq) M2 EITHER acid-base reaction of water by ammonia OR dissociation of (some) Mg(OH) ₂	2						

20. M/J 24/P21/Q3(a)

3(a)	M1 (if a change in conditions occurs) the equilibrium shifts M2 to minimise the change in conditions	2
------	---	----------

21. M/J 24/P21/Q3(b)-(c)

3(b)	M1 change in appearance paler red / more yellow / less orange M2 change in relative concentration of $\text{FeSCN}^{2+}(\text{aq})$ lower M3 change in value of the equilibrium constant, K_c lower	3
3(c)(i)	M1 deduce the concentration of SCN^- present in initial mixture $[\text{SCN}^- \text{ eq'm}] = [\text{Fe}^{3+} \text{ eq'm}] = [\text{original SCN}^-] - [\text{FeSCN}^{2+} \text{ produced}]$ $1.30 \times 10^{-3} + 0.300 \times 10^{-3} = 1.60 \times 10^{-3} \text{ mol dm}^{-3}$ M2 find initial amount of Fe^{3+} initially $\text{M1} \times 25 / 1000 = \text{amount of Fe}^{3+} \text{ added initially} = 4.00 \times 10^{-5} \text{ (mol)}$	2
3(c)(ii)	M1 value for $K_c = 178$ M2 units = $\text{mol}^{-1} \text{ dm}^3$	2

22. M/J 24/P22/Q3(a)

3(a)	M1 rates of forward and reverse / backward reactions are equal M2 no change in measurable properties OR concentration of reactants AND products remain constant	2
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23. M/J 24/P22/Q3(b)

3(b)(i)	M1 change to appearance of the mixture: (goes) darker red / more red M2 change to relative concentration of $\text{Fe}^{3+}(\text{aq})$: decreases M3 change to the value of equilibrium constant, K_c : constant / none	3
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3(b)(ii)	Method 1 M1 conc of initial conc of Fe^{3+} and $\text{SCN}^- = 0.0020 \text{ (mol dm}^{-3}\text{)}$ M2 at equilibrium the conc of Fe^{3+} and SCN^- are $(\text{M1} - 0.000423)$ M3 use $0.000423 \div (\text{M1} - 0.000423)^2$ to calculate $K_c = 170(.089)$ M4 units = $\text{mol}^{-1} \text{ dm}^3$ OR $\text{dm}^3 \text{ mol}^{-1}$ Method 2 M1 no mol $\text{FeSCN}^{2+} = 4.23 \times 10^{-4} \times 25 / 1000 = 1.0575 \times 10^{-5}$ M2 no mol Fe^{3+} and SCN^- are $(5 \times 10^{-5} - \text{M1}) = 3.9425 \times 10^{-5}$ M3 $0.000423 \div (\text{M2} \times 1000 / 25)^2 = 170(.089)$ M4 units = $\text{mol}^{-1} \text{ dm}^3$ OR $\text{dm}^3 \text{ mol}^{-1}$	4
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24. M/J 24/P23/Q2(a)

2(a)	58% / no change / stays the same AND catalysts don't affect yield / iron is a catalyst	1
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25. M/J 24/P23/Q2(b)(i)

2(b)(i)	less yield (in C than A) AND (as) exothermic (forward reaction)	1
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26. M/J 24/P23/Q2(c)

2(c)(i)	M1 $K_p = (\text{pNH}_3)^2 / (\text{pN}_2)(\text{pH}_2)^3$ M2 atm^{-2}	2
2(c)(ii)	M1 correct amount N_2 at equilibrium = $1 - 1.16 / 2 = 0.42$ AND correct amount H_2 at equilibrium = $3.0 - (1.16 \times 3 / 2) = 1.26$ M2 express amount of ammonia as a mol fraction in terms of total amount of $\text{N}_2 + \text{H}_2 + \text{NH}_3$ at equilibrium = $1.16 / (\text{N}_2 + \text{H}_2 + 1.16) = 0.408$ OR 0.41	2
2(c)(iii)	$0.625 \times 1000 = 625 \text{ (atm)}$	1

27. M/J 24/P23/Q4(b)(i)

4(b)(i)	acid–base reaction	1
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28. O/N 23/P22/Q2(a)

2(a)	curved line from (0,0) to reach same horizontal line	1
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29. O/N 23/P22/Q2(b)

2(b)	equilibrium position would / it moves to left / reactants	1
	forward reaction is exothermic (so disfavoured by increase in T)	1

30. O/N 23/P22/Q2(c)

2(c)	$\frac{0.605}{1.60}$ (= 0.378125) $\frac{0.605}{1.60}$ (= 0.378125) $\frac{0.390}{1.60}$ (= 0.24375)	1
	(partial pressure / kPa 38.19 38.19 24.62)	1
	$K_p = 24.62 \div 38.19^2 = 0.0169$	
	kPa^{-1}	1

31. O/N 23/P22/Q3(a)(ii)

3(a)(ii)	proton / H^+ donor	1
	partially dissociates (in solution / in water / when aqueous)	1

32. M/J 23/P22/Q4(b)

4(b)	M1 amount ethanol increases M2 equilibrium moves to right AND to produce fewer moles gas OR smaller amount of gas (to reduce the pressure)	2
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33. M/J 23/P22/Q4(c)(i)

4(c)(i)	acid AND it donates a proton / H^+	1
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34. M/J 23/P23/Q4(b)

4(b)(i)	(from colourless) to purple	1
4(b)(ii)	$K_c = [H_2][I_2] / [HI]^2$ OR $0.0217 = [H_2][I_2] / [HI]^2$	1
4(b)(iii)	M1 $[HI]$ at equilibrium = $1.7 / 2 = 0.85$ M2 idea that $[H_2] = [I_2]$ in equilibrium mixture M3 correct calculation based on correct rearrangement of equation in (b)(ii), 0.0217 and their M1 = 0.125	3

35. M/J 23/P23/Q4(c)

4(c)(i)	new mixture will have a lower / smaller conc / amount of H_2 and I_2	1
4(c)(ii)	endothermic AND decrease in temperature causes position of equilibrium to shift in exothermic direction	1

36. MAR 23/P22/Q3(d)(ii)

3(d)(ii)	equilibrium moves to right/products where fewer moles / smaller amount of gas	1
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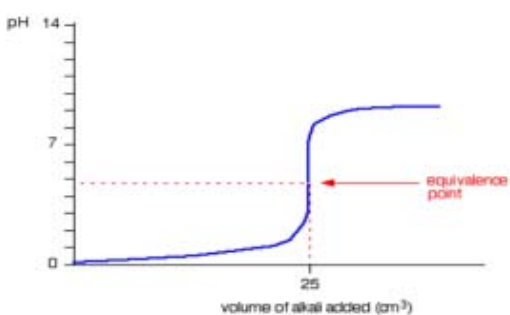
37. O/N 22/P22/Q1(c)(i)

1(c)(i)	proton / H^+ donor	1
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38. O/N 22/P22/Q2(c)(ii)

2(c)(ii)	reaction is exothermic so equilibrium shifts to the LHS OR equilibrium shifts to LHS because forward reaction is exothermic OR equilibrium shifts to LHS because backward reaction is endothermic OR endothermic reaction is favoured so more reactants / are $AlCl_3$ is made	1
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39. M/J 22/P21/Q2(d)

2(d)(i)	M1 proton / H^+ donor M2 fully dissociates (in aqueous solution / water / solvent)	2
2(d)(ii)	$NH_3 + H_2O \rightleftharpoons NH_4^+ + OH^-$	1
2(d)(iii)	M1 correct basic shape extending to $\sim 50\text{ cm}^3$ with vertical portion of curve at 25 cm^3 M2 initial pH at 0–2 (based on idea that HCl is a strong acid) AND final pH at between 8–12 (based on idea that NH_3 is a weak alkali) 	2

40. M/J 22/P22/Q3(a)

3(a)(i)	M1 rate(s) of forward and reverse / backward reactions are equal / are the same M2 no change in measurable properties OR concentration of reactants and products remain constant	2
3(a)(ii)	(colourless gas) becomes purple (gas)	1
3(a)(iii)	$K_p = \frac{(pH^2)(pI^2)}{(pHI)^2}$	1
3(a)(iv)	M1 use correct expression for finding amount of HI used up in reaction $86.4 = 100 \times (0.025 - x) / 0.025 - x + x / 2 + x / 2$ where $x = HI \text{ mol used up}$ OR $86.4 = 100 \times (0.025 - 2x) / (0.025 - 2x + x + x)$ where $2x = HI \text{ mol used up}$ M2 correct calculation to find amount of HI(mol) at equilibrium based on $HI(mol) \text{ at eq'm} = 0.025 - HI(mol) \text{ used} = 0.025 - 0.0034 = 0.0216$ OR 0.022 (mol) alternative method M1 use the pp H_2 and pp I_2 at equilibrium (= 13.6) to calculate no mol H_2 & I_2 at equilibrium (either 0.0034 in total OR 0.0017 each) M2 use value for no mol $H_2 + I_2$ in expression to find no mol HI $0.025 - (\text{no mol } H_2 + I_2) = 0.0216$	2

41. M/J 22/P22/Q3(c)

3(c)	none / no change	1
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42. M/J 22/P23/Q3(a)(iii)

3(a)(iii)	M1 $K_c = \frac{[C_2H_6][CO][CH_4]}{[C_4H_{10}O]}$ M2 $\text{mol}^2 \text{dm}^{-6}$	2
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43. MAR 22/P22/Q2(d)(iii)

2(d)(iii)	position of equilibrium moves / farther to right (at 20 km) [1] (forward) reaction is exothermic AND temperature colder at 20 km (cf. 50 km) [1]	2
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44. MAR 22/P22/Q3(d)

3(d)(i)	proton / H^+ donor [1] fully dissociates (in aqueous solution / water / solvent) [1]	2
3(d)(ii)	M1: correct sigmoid shape with vertical section at 25 cm^3 for both M2: both curves show initial $\text{pH} \leq 2$ M3: (with NaOH) heading to $\text{pH} \geq 12$ (with NH_3) heading to $\text{pH} 8-12$ 	3